Talha Rashid

Entry-Level Data Scientist

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Profile

Data Science undergraduate focused on NLP, retrieval augmented generation, model fine-tuning, and practical MLOps. Comfortable moving from messy data or model questions to reproducible Python experiments, grounded LLM prototypes, deployment ready APIs, and clear technical documentation.

Technical Skills

Languages	Python, SQL, JavaScript/TypeScript, LaTeX
ML / Data	PyTorch, scikit-learn, pandas, NumPy, RDKit, MLflow, model evaluation, feature engineering
NLP / LLMs	RAG, LangChain, Qdrant, embeddings, PubMedBERT, BioMistral, QLoRA, LoRA adapters, DeepSpeed ZeRO-2
MLOps / APIs	Git, GitHub Actions, Docker, FastAPI, Flask, pytest, Prometheus, Grafana, joblib
Analytics	Search-query analysis, CTR/CVR/ACOS diagnostics, KPI reporting, recommendation design, A/B test planning

Education

FAST NUCES 2021–2025
BS Data Science *Undergraduate*

- Relevant coursework: Machine Learning, Natural Language Processing, Statistics, Data Structures, Databases, Linear Algebra.

Beaconhouse School System, Margalla Campus O-level–A-level

Certifications

Perform Cloud Data Science with Azure Machine Learning – Microsoft Jun 2024–Jan 2033
Google Data Analytics Certificate – Google

Projects

Applied LLM Systems: SkinSync RAG + Fine-Tuning Benchmark *LangChain, Qdrant, PubMedBERT, BioMistral, QLoRA, DeepSpeed*

- Built SkinSync, a skincare RAG chatbot prototype using FastAPI, LangChain RetrievalQA, Qdrant vector search, PubMedBERT embeddings, and a local quantized BioMistral GGUF model.
- Implemented PDF ingestion with chunking, 768-dimensional biomedical embeddings, cosine similarity retrieval, source aware responses, and prompts designed to avoid unsupported medical claims.
- Compared code-model fine-tuning strategies on Qwen2.5-0.5B-Instruct using CodeAlpaca: single-GPU QLoRA on a Tesla T4 versus dual-GPU DeepSpeed ZeRO-2 on RTX A4000 hardware.
- Reported trade-offs across trainable parameters, runtime, memory, and evaluation loss, including 8.80M adapter parameters versus 494.03M full parameters.

Tox21 Molecular Toxicity MLOps Platform *FastAPI, Docker, MLflow, Prometheus, Grafana, GitHub Actions*

- Built an end-to-end MLOps platform for Tox21 SR-ARE toxicity prediction using RDKit Morgan fingerprints, multi-model training, and ROC-AUC based model selection.
- Compared Logistic Regression, Random Forest, and Extra Trees models, selected Extra Trees with 0.7648 ROC-AUC, and served predictions through a FastAPI endpoint.
- Added Docker Compose orchestration, Prometheus metrics, Grafana dashboarding, API tests, CI validation, and GHCR image publishing through GitHub Actions.

Similarity GAN for CIFAR-10 Cats and Dogs *PyTorch, torchvision, DCGAN, Siamese Networks*

- Built a custom GAN training pipeline using a DCGAN-style generator and a Siamese similarity discriminator trained on paired-image comparisons.
- Designed the discriminator around shared convolutional embeddings and absolute embedding differences for real-real versus real-generated pairs.
- Filtered CIFAR-10 to cat/dog classes, configured reproducible dataloaders, and saved epoch-level checkpoints and generated image samples for inspection.

Amazon ASIN Search Performance Assessment *Data Analysis, Ads KPIs, Experiment Planning*

- Analyzed paid search and Search Query Performance patterns across three Amazon ASINs, separating visibility, click through, conversion, and profitability issues for each product.
- Identified the strongest ASIN using 60-day ad metrics, including 1.19% CTR, 11.07% CVR, and 23.36% ACOS, then proposed exact-match scaling, waste pruning, and listing tests.
- Wrote recommendations for keyword segmentation, negative targeting, listing rewrites, image tests, and experiment KPIs across three product categories.